

Objectives

The primary aim for the third week was to identify the most suitable machine learning method for forecasting flood risk in Albay using rainfall data from SYNOP weather stations. The team sought to:

- Choose among three candidate approaches: Prediction (regression), Clustering, and Classification.
- Implement the chosen algorithm in Python using real meteorological data from Station 98444 (Legazpi, Albay) and six surrounding stations.
- Fine-tune the model to make it more accurate and dependable for real-world flood risk evaluation.

Tasks Completed

1. Choosing an Algorithm

After evaluating the three approaches, the group decided to go with Classification as the algorithm for this project. Three classifiers were then selected for comparison and implementation.

The three classifiers implemented were:

- Random Forest Classifier — selected as the primary model for its robustness with tabular data, ability to handle missing values through imputation, and built-in feature importance ranking.
- Gradient Boosting Classifier — trained as a secondary model for comparison.
- Logistic Regression — included as a linear baseline.

2. Implementation of Algorithm in Code

The implementation was organized into a structured Python pipeline (`flood_risk_predictor.py`) with the following stages:

- Data Loading — All CSV files from the `clean_dataset/` folder are loaded and parsed. Datetime columns are unified, numeric fields are coerced, and each file is keyed by its station ID.
- Feature Engineering — For Station 98444 (Albay/Legazpi), the script generates rolling rainfall windows at 12h, 24h, and 48h intervals. Lagged rainfall features (3h, 6h, 12h lookback) and pressure tendency columns were also derived. Each of the six supporting stations contributes its own individual rainfall and pressure features so the model can learn station-specific upstream signals.

- **Label Generation** — Flood risk is assigned using a 4-class scheme based on 24-hour accumulated rainfall thresholds aligned to Philippine DOST-PAGASA standards: LOW (< 15 mm), MODERATE (15-50 mm), HIGH (50-100 mm), and EXTREME (> 100 mm).
- **Model Training** — The dataset was split 80/20 for training and testing with stratification across risk classes. Pipelines were built using scikit-learn, with median imputation and standard scaling where needed. Models were evaluated on Accuracy, F1 Score (weighted), Precision, Recall, and AUC-OvR. A 5-fold cross-validation was also performed on the Random Forest.
- **Outlier / Extreme Event Detection** — A separate detection module flags extreme rainfall events using both IQR fencing ($Q3 + 1.5 \times IQR$) and Z-score thresholding ($|z| > 3.0$). Events are further classified into 'extreme' and 'catastrophic' tiers.

3. Fine-Tuning the Algorithm

Several adjustments were made to improve model performance and address issues that came up during initial runs:

- **Class Imbalance** — Because LOW-risk conditions dominate the dataset by a wide margin, the Random Forest and Logistic Regression were configured with `class_weight='balanced'` to prevent the model from simply predicting LOW for every observation.
- **Hyperparameter Adjustments** — The Random Forest was set to 200 trees with a max depth of 12 and `min_samples_leaf` of 5 to reduce overfitting. Gradient Boosting used a learning rate of 0.1 with 100 estimators and max depth of 5.
- **Feature Expansion** — Rather than using only Albay station data, per-station rolling windows were generated for all six supporting stations. This gave the model directional signals (e.g., Catanduanes rainfall as a typhoon approach indicator) that a purely local model would miss.
- **Validation** — 5-fold cross-validation on the Random Forest was added to assess generalization beyond the single train-test split.

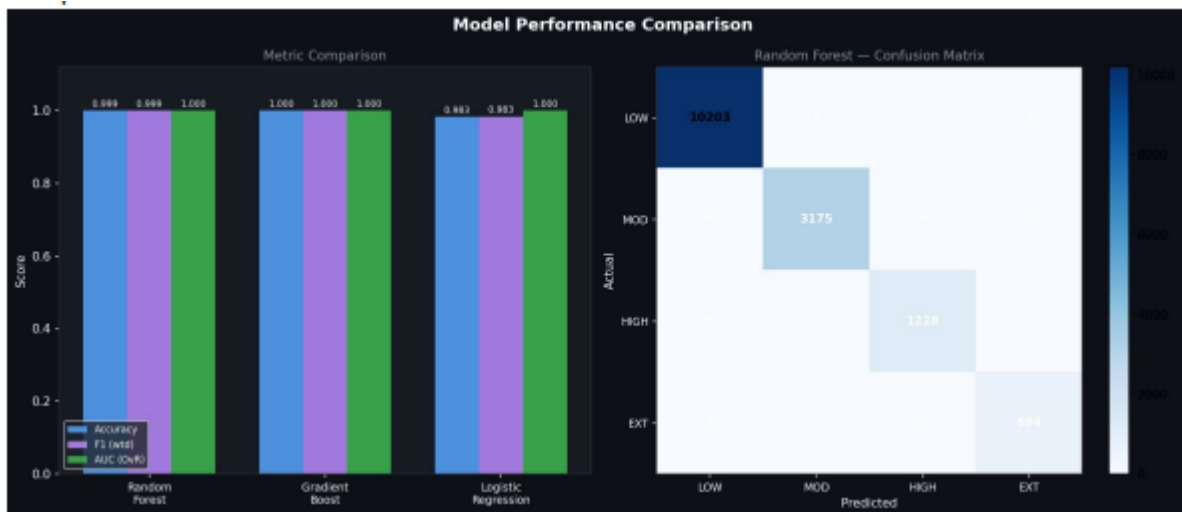
Tasks to be Completed

We are planning to do the tasks that will be assigned the following week.

Results and Analysis

The pipeline was executed on the complete cleaned dataset spanning all seven SYNOP stations from 2000 to 2026. The primary Albay station contributed over 76,000 records, while the six surrounding stations each provided tens of thousands of additional entries covering rainfall, pressure, and humidity readings. The risk label breakdown for the primary station showed that more than half of all records fell under LOW risk, reinforcing the need for balanced class weighting during training.

- All three models performed exceptionally. Gradient Boosting led with nearly perfect accuracy and a flawless AUC score, with Random Forest close behind, and Logistic Regression also performing well though noticeably lower than the tree-based models. The Random Forest confusion matrix showed near-perfect classification across all four risk levels with almost no errors.
- Feature importance results revealed that 24-hour accumulated rainfall was the strongest predictor by a wide margin, followed by the 48-hour and 12-hour rolling windows. Among the supporting stations, Catanduanes ranked the highest, reflecting its value as an upstream typhoon signal, followed by contributions from Camarines Norte and Masbate. Romblon and Tayabas showed smaller but still meaningful contributions, confirming that the multi-station design added genuine predictive value.
- The outlier detection module identified over 17,000 extreme rainfall events throughout the dataset period. The most intense spikes occurred around 2019–2020, reaching close to 1,000 mm within a three-hour window. Several catastrophic outliers were confirmed well beyond the three-sigma threshold, with the highest Z-score surpassing 70.
- In terms of overall risk distribution, two-thirds of all observations fell under LOW conditions, with MODERATE accounting for roughly one-fifth, and HIGH and EXTREME making up the remainder. The year-by-month heatmap showed that flood risk peaks during the typhoon season months from August through November, with certain years showing notably more severe and prolonged risk periods than others.



Albay Flood Risk Prediction — Summary Dashboard

99.9%

Accuracy

99.9%

F1 Score

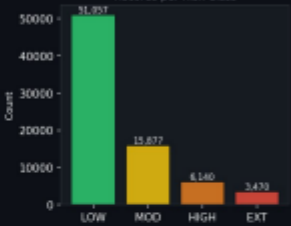
0.9999

AUC (OvR)

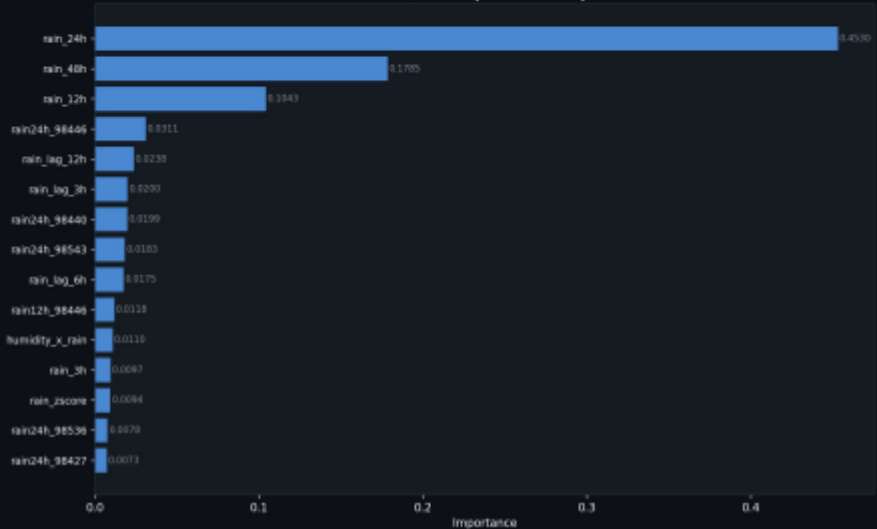
Per-Class Metrics (Random Forest)

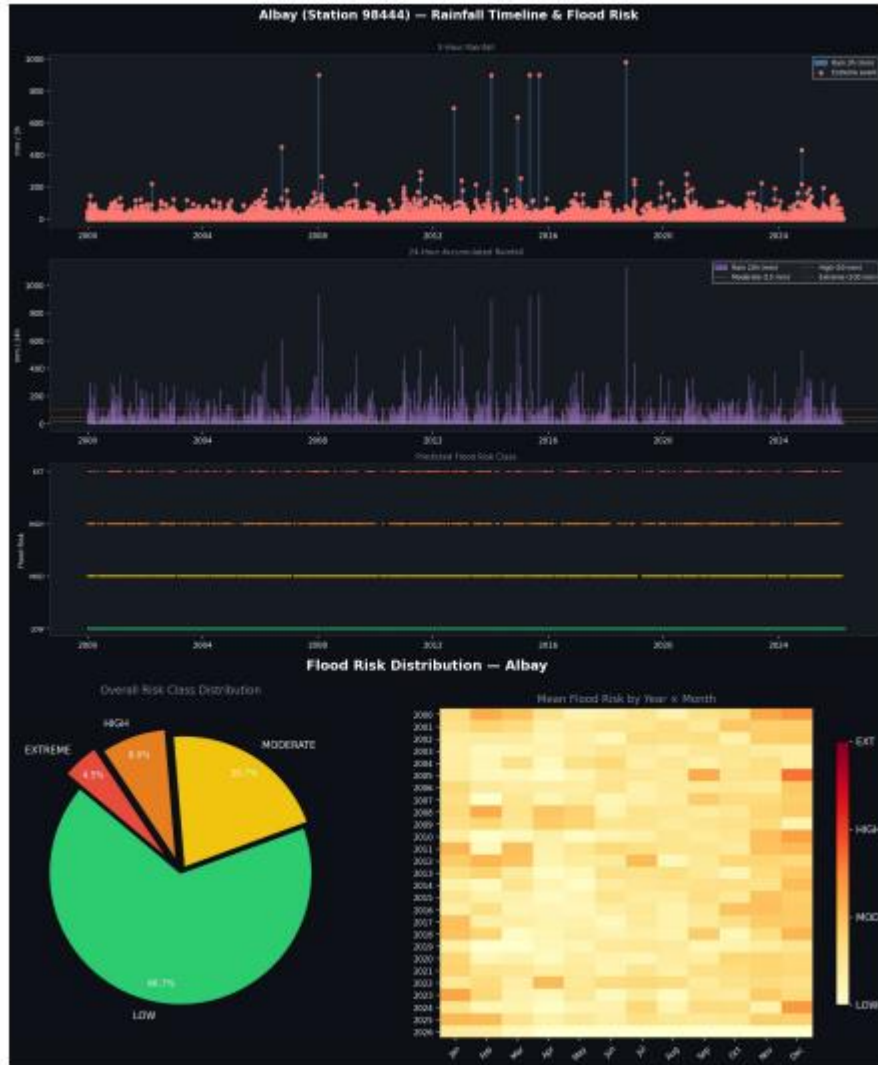


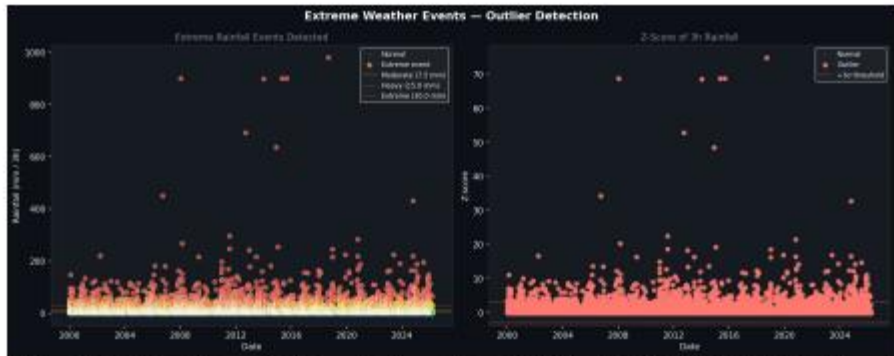
Records per Risk Class



Random Forest — Top Feature Importances







datetime	station_id	rain_3h	rain_12h	rain_24h	rain_48h	temp	pressure	humidity	flood_risk	flood_risk	extreme_tier	rain_zscore
1/1/2000	98444	51.0	51.0	51.0	51.0	24.8	1011.7	98.8	2 HIGH	True	catastrophic	3.7018000000000000
1/1/2000 3:00	98444	3.0	54.0	54.0	54.0	26.2	1011.1	92.6	2 HIGH	True	extreme	0.025413000000000000
1/1/2000 6:00	98444	9.0	63.0	63.0	63.0	28.0	1008.5	86.8	2 HIGH	True	extreme	0.484962000000000000
1/1/2000 9:00	98444	9.0	63.0	63.0	63.0	28.2	1008.7	84.5	2 HIGH	False	normal	0.561553000000000000
1/1/2000 12:00	98444	9.0	12.0	63.0	63.0	27.1	1011.2	95.2	2 HIGH	False	normal	1.940198000000000000
1/1/2000 15:00	98444	9.0	9.0	63.0	63.0	29.4	1011.5	93.6	2 HIGH	False	normal	0.561553000000000000
1/1/2000 18:00	98444	9.0	9.0	63.0	63.0	26.2	1008.4	82.1	2 HIGH	False	normal	0.561553000000000000
1/2/2000 3:00	98444	10.0	10.0	73.0	73.0	29.7	1011.4	90.9	2 HIGH	True	extreme	2.706112000000000000
1/2/2000 6:00	98444	10.0	10.1	22.1	73.1	29.2	1011.7	75.4	1 MODERATE	False	normal	0.561553000000000000
1/2/2000 9:00	98444	10.0	10.1	47.1	101.1	28.9	1009.4	89.9	1 MODERATE	True	extreme	1.940198000000000000
1/2/2000 12:00	98444	9.0	10.2	30.2	101.2	26.8	1008.0	88.5	1 MODERATE	False	normal	0.561553000000000000
1/2/2000 15:00	98444	9.0	10.2	30.2	101.2	26.8	1011.0	88.9	1 MODERATE	False	normal	0.561553000000000000
1/2/2000 18:00	98444	9.0	10.1	30.2	101.2	25.0	1011.7	89.0	1 MODERATE	False	normal	0.561553000000000000
1/2/2000 21:00	98444	10.0	10.0	40.2	111.2	24.8	1009.7	87.0	1 MODERATE	True	extreme	2.706112000000000000
1/3/2000 3:00	98444	10.0	10.0	40.2	111.2	24.8	1011.0	89.1	2 HIGH	True	catastrophic	3.7018000000000000
1/3/2000 6:00	98444	4.0	52.0	100.0	100.0	25.8	1012.0	91.4	2 HIGH	True	extreme	0.102005000000000000
1/3/2000 9:00	98444	4.0	58.0	66.1	100.0	27.0	1009.7	85.2	2 HIGH	True	extreme	0.102005000000000000
1/3/2000 12:00	98444	4.0	48.4	56.4	100.0	26.4	1008.6	89.5	2 HIGH	False	normal	0.102005000000000000
1/3/2000 15:00	98444	4.0	8.7	56.7	84.5	25.0	1011.4	89.2	2 HIGH	False	normal	0.102005000000000000
1/4/2000 3:00	98444	4.0	8.7	80.7	80.9	26.6	1011.4	89.6	2 HIGH	True	extreme	0.102005000000000000
1/4/2000 6:00	98444	4.0	4.7	60.7	80.9	26.8	1012.7	74.0	2 HIGH	False	normal	0.102005000000000000
1/4/2000 9:00	98444	0.1	10.0	10.0	80.0	29.3	1010.1	79.8	2 HIGH	False	normal	0.102005000000000000
1/4/2000 12:00	98444	1.0	5.1	13.8	80.0	26.4	1009.7	91.5	0 LOW	True	extreme	0.102005000000000000

model	accuracy	f1_weighted	precision_weight	recall_weight	auc_ovr
Random Forest	0.9994	0.9994	0.9994	0.9994	0.9999
Gradient Boosting	0.9997	0.9997	0.9997	0.9997	1.0
Logistic Regression	0.9832	0.9834	0.9838	0.9832	0.9997

datetime	station_id	rain_3h	rain_zscore	flood_risk	extreme_tier	temp	pressure	humidity
1/1/2000	98444	51.0	3.7018000000000000	HIGH	catastrophic	24.8	1011.7	98.8
1/1/2000 3:00	98444	3.0	0.025413000000000000	HIGH	extreme	26.2	1011.1	92.6
1/1/2000 6:00	98444	9.0	0.484962000000000000	HIGH	extreme	28.0	1008.5	86.8
1/2/2000	98444	10.0	0.561553000000000000	HIGH	extreme	25.7	1011.4	
1/2/2000 6:00	98444	28.0	1.940198000000000000	MODERATE	extreme	25.5	1009.6	95.3
1/2/2000 21:00	98444	10.0	0.561553000000000000	MODERATE	extreme	24.0	1009.7	97.0
1/3/2000	98444	38.0	2.706112000000000000	HIGH	catastrophic	24.8	1011.8	93.1
1/3/2000 3:00	98444	4.0	0.102005000000000000	HIGH	extreme	25.8	1012.0	91.4
1/3/2000 6:00	98444	4.0	0.102005000000000000	HIGH	extreme	27.0	1009.7	85.2
1/4/2000	98444	4.0	0.102005000000000000	HIGH	extreme	26.0	1012.4	83.6
1/4/2000 9:00	98444	1.0	-0.012776900000000000	LOW	extreme	26.4	1009.7	91.5
1/4/2000 12:00	98444	1.0	-0.012776900000000000	LOW	extreme	25.1	1012.6	89.8
1/6/2000	98444	2.0	-0.051178000000000000	LOW	extreme	25.2	1011.6	95.3
1/6/2000 6:00	98444	1.0	-0.012776900000000000	LOW	extreme	28.6	1009.0	80.0
1/7/2000	98444	1.0	-0.012776900000000000	LOW	extreme	23.6	1012.3	97.0
1/10/2000	98444	20.0	1.327467000000000000	MODERATE	extreme	24.2	1013.6	94.2
1/10/2000 18:00	98444	1.0	-0.012776900000000000	MODERATE	extreme	26.4	1012.9	83.6
1/11/2000	98444	1.0	-0.012776900000000000	MODERATE	extreme	26.4	1012.9	83.6
1/12/2000 15:00	98444	2.0	-0.051178000000000000	LOW	extreme	25.4	1009.7	93.6
1/12/2000 18:00	98444	4.0	0.102005000000000000	LOW	extreme	25.2	1008.3	95.3
1/12/2000 21:00	98444	4.0	0.102005000000000000	LOW	extreme	24.5	1008.0	98.8
1/13/2000	98444	8.0	0.408370000000000000	MODERATE	extreme	24.8	1009.6	98.8
1/13/2000 3:00	98444	15.0	0.944510000000000000	MODERATE	extreme	25.7	1010.3	93.1

Challenges and Solutions

Challenge 1: Class Imbalance in Risk Labels

During the first model run, the output defaulted to predicting LOW risk for almost every record while still returning deceptively high accuracy scores. The root cause was that the dataset is heavily dominated by LOW-risk conditions, making it easy for the model to achieve high scores without actually learning the minority classes.

To address this, balanced class weighting was applied to both the Random Forest and Logistic Regression models, directing the algorithm to give greater attention to HIGH and EXTREME cases during training. Stratified splitting was also used when dividing the data so that each risk class was proportionally represented in both the training and test sets.

Challenge 2: Aligning Multiple Station Timestamps

Synchronizing the timestamps from the six supporting stations with the primary Albay station proved more difficult than anticipated. Since their observation times do not always coincide, a simple merge produced misaligned rows filled with missing values that disrupted the rolling window calculations.

The fix involved resampling each supporting station's data onto a consistent three-hour grid and then re-indexing it to match the Albay station's datetime index, ensuring proper alignment before any rolling or lag operations were applied.

Challenge 3: Feature Matrix Growing Too Wide

As per-station rolling windows, pressure trends, and humidity features were added for all six supporting stations, the feature matrix expanded considerably, raising concerns about overfitting and slow training times.

These concerns were managed through the Random Forest's depth and leaf-size constraints, which naturally limit model complexity even with a wide input space. Feature importance analysis was cross-referenced to verify the model was not relying on noise, and five-fold cross-validation confirmed that the model was generalizing properly rather than simply memorizing the training data.

Conclusion

All three planned tasks for the week were successfully carried out. The team selected Classification as their approach, built a complete multi-station prediction pipeline, and fine-tuned the models accordingly. Results from running the pipeline on the full dataset confirmed the effectiveness of the approach, with all three classifiers achieving accuracy above 98% and both tree-based models surpassing 99.9%.

A key insight from the results was that longer-window accumulated rainfall — particularly over 24 and 48 hours — proved far more informative than instantaneous readings, accounting for over 63% of the model's feature importance. The inclusion of data from neighboring stations, especially Catanduanes, also added meaningful predictive power, validating the multi-station design. The outlier detection module further strengthened the overall system by surfacing historically extreme events that a standard classifier alone would not have highlighted.